

INTRODUCTION TO ENGINEERING • LAB 3

Systems Thinking II

Reading and Drawing Process Flow Diagrams

A recap of Lecture 3, followed by small-group practice: opening the box, naming the symbols, and drawing a PFD in box language.

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What a Process Flow Diagram Shows

A process flow diagram (PFD) shows the steps of a process in order, and what flows between them.

Sequence

What happens first, next, and last.

Flow

What moves between steps: material, people, information, money, energy.

Problems

Gaps, waits, loops, and hand-offs become visible.

A PFD is a model of a process, drawn at one chosen level of detail.

The Building Blocks — Five Symbols



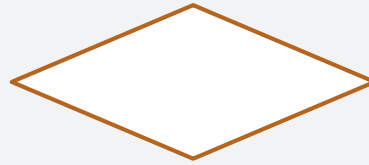
Start / End

Where the process begins and finishes



Step

One action, named verb + noun



Decision

A question with labeled exits:
Yes / No



Flow

Order of steps, and what moves



Wait / queue

Where things sit and wait

Five symbols are enough for this course. Consistency matters more than which symbols you choose.

LECTURE RECAP

Levels of Detail, and What Arrows Carry

LEVEL 0

Black box

Inputs go in, outputs come out. Nothing inside is shown.

LEVEL 1

Main steps

The handful of major steps inside the boundary.

LEVEL 2

Zoom in

One Level 1 step opened up into its own diagram.

Every arrow carries something:

Material · People · Information · Money · Energy

Choose the level that fits your question, and keep one level per diagram.

Five Rules for a Readable Diagram

- 1** Start and end clearly — one start, and every path reaches an end.
- 2** Name steps with a verb + noun — “Receive deliveries”, not “Delivery”.
- 3** Label the arrows — say what moves: material, people, information, money.
- 4** Give every decision its exits — each Yes and each No must lead somewhere.
- 5** Keep one level of detail — arrows crossing the boundary are your inputs and outputs.

LAB ACTIVITY

Small-Group Practice

Open the box, agree on the boundary, and draw the process in box language.

Your Task

Task:

- Define the system — in one sentence, say what it is and what it does.
- Draw the boundary — decide, and state, what is inside and what is outside.
- Map inputs and outputs — list what crosses the boundary, in and out.
- Identify stakeholders — who Uses It, who Runs It, who is Affected By It.
- Note the operating context — what constrains the system from just outside the boundary.
- Propose one solution — one concrete change you would make, and why.

Groups & time

Teams of 3–4. 15 minutes to work the task, 5 minutes to share one diagram per group.

Deliverable

A Level 0 black box and a Level 1 box-language diagram, on paper or whiteboard — labels on every arrow.

EXAMPLE SYSTEM

1 · Solar Panel Assembly Line

- 1
 - Each day 600 glass sheets and 600 backsheets are received, together with 36,000 solar cells, EVA film, aluminium frames and junction boxes.
 - Glass and other materials are inspected, cells are sorted and stringed, then modules are laid up with EVA and backsheets, laminated and sent to a cooling queue for 30 minutes.
 - Panels are trimmed, framed and junction boxes installed. Each module is flash tested and inspected by electroluminescence, then decided either pass, or go to rework or scrap.
 - Approved modules are packed, and the finished panels produced each day are shipped to customers.

Task:

define system, draw boundary, map inputs/outputs, identify Uses It / Runs It / Affected By It, note operating context, propose one solution.

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1

Solar Panel Assembly Line



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Task: define system, draw boundary, map inputs/outputs, identify Uses It / Runs It / Affected By It, note operating context, propose one solution.

WORKED SOLUTION

A) System, Boundary & Stakeholders

System & boundary

- System: the solar panel assembly line, from receiving dock to shipping dock.
- Inside the boundary: inspection, stringing, lay-up, lamination, cooling, trimming/framing, testing, and packing.
- Outside the boundary: raw-material suppliers, the customer, and the waste/recycling handler.

Stakeholders

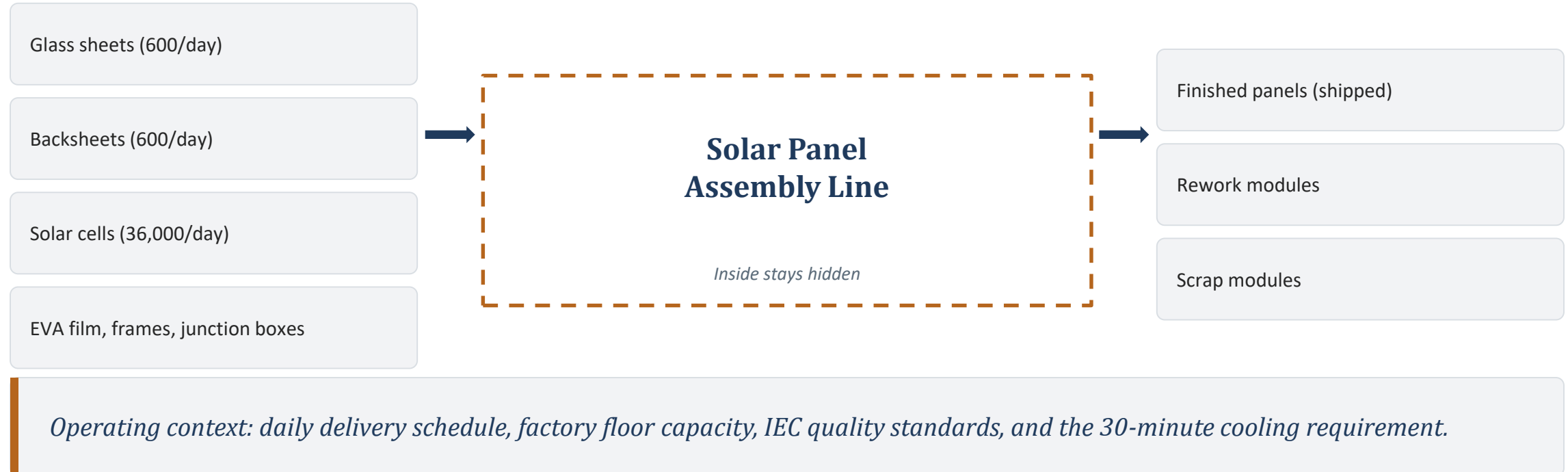
- Uses It: the customer who receives finished panels.
- Runs It: line operators, quality inspectors, and the shift supervisor.
- Affected By It: maintenance crew, logistics team, and the environment (scrap and waste streams).

Inputs / Outputs

In: 600 glass sheets/day, 600 backsheets/day, 36,000 cells/day, EVA film, aluminium frames, junction boxes. Out: finished panels (shipped), rework modules, scrap modules.

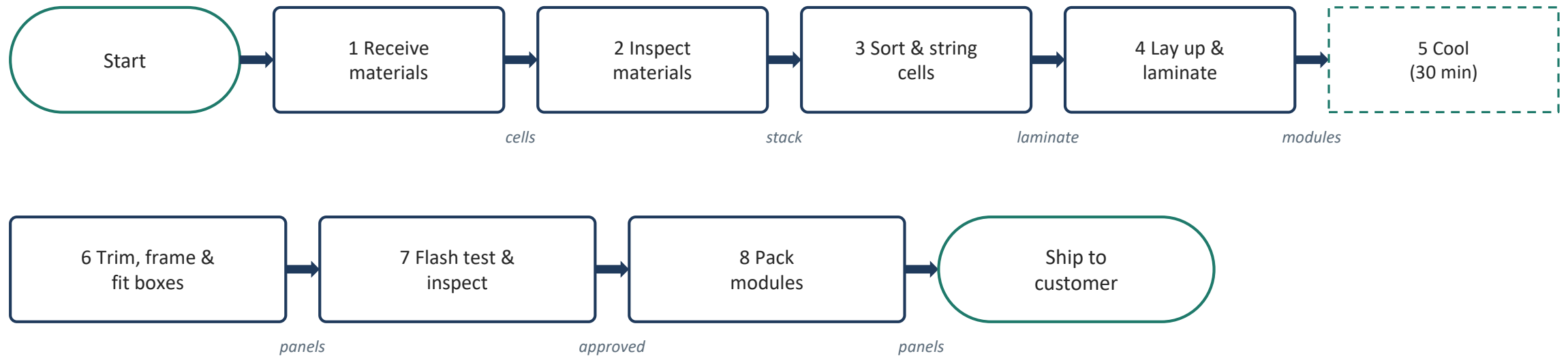
WORKED SOLUTION

B) Level 0 — The Black Box



WORKED SOLUTION

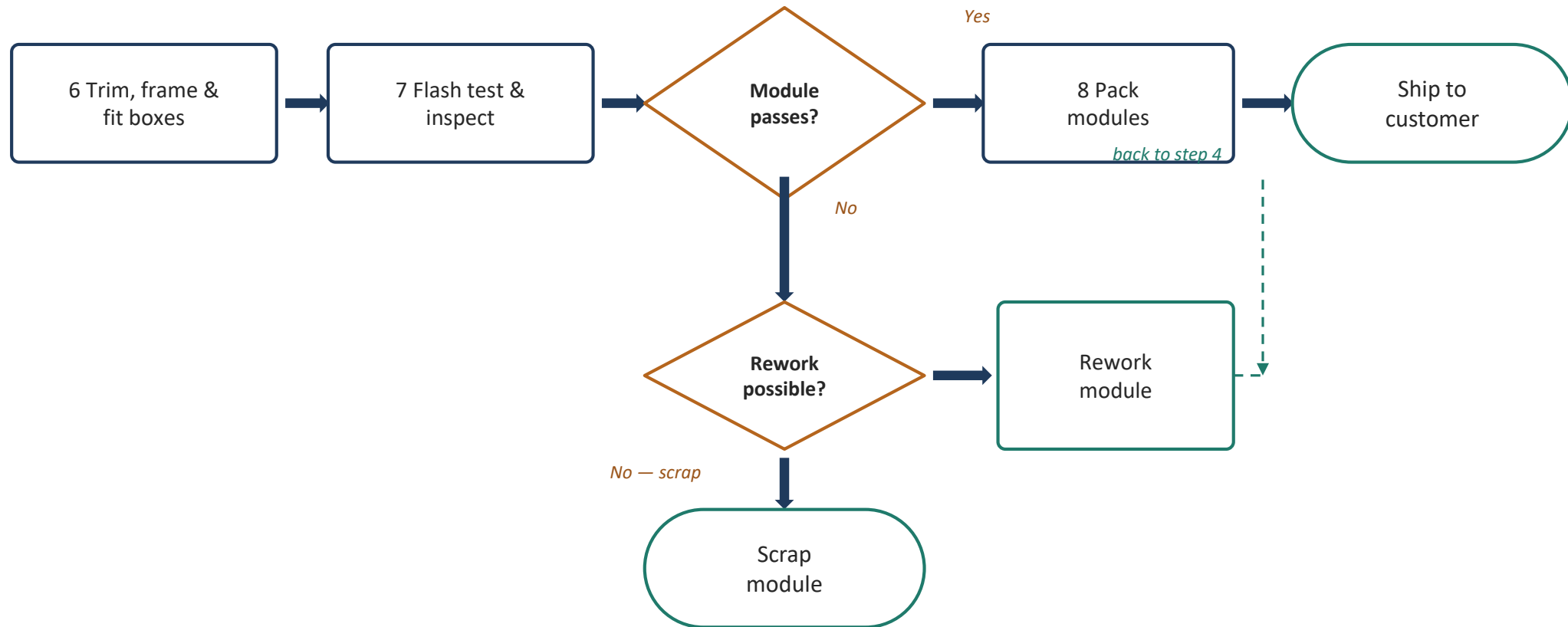
C) Level 1 — The Main Flow (Box Language)



A dashed box marks a wait. Step 7's decision (pass / rework / scrap) is zoomed into on the next slide.

WORKED SOLUTION

D) Adding the Decision: Test Result



Every decision gets both exits a destination. Rework loops back upstream; scrap exits the system as waste (This example solution and it can vary according to how you understand the logic).

Thank You

Questions on today's activity, or on drawing PFDs for your own project, are welcome any time.

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